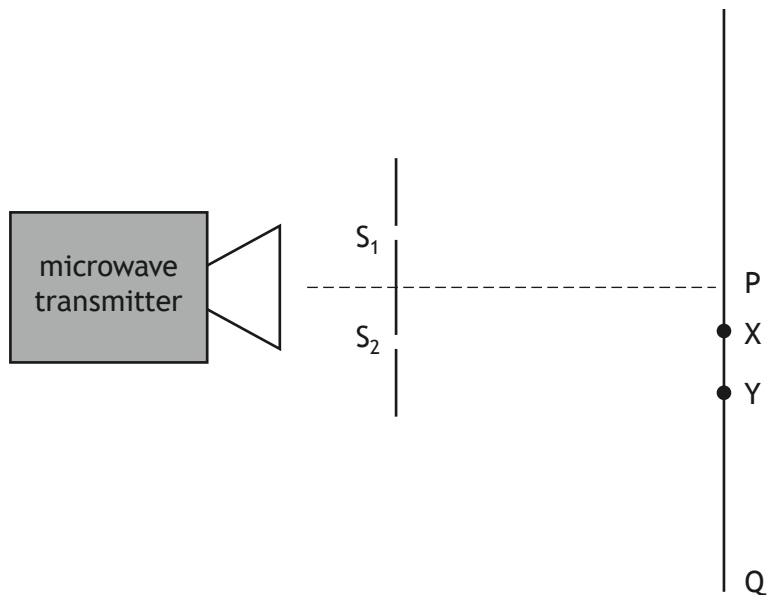


16. A student sets up the apparatus shown to produce an interference pattern. S_1 and S_2 provide coherent sources of microwave radiation.



A radiation detector is moved along the line from P to Q.

The first minimum is detected at point X.

The distance from S_1 to X is 300 mm.

The distance from S_2 to X is 250 mm.

The second minimum is detected at Y.

The path difference $S_1Y - S_2Y$ is

- A 25 mm
- B 75 mm
- C 100 mm
- D 150 mm
- E 250 mm.